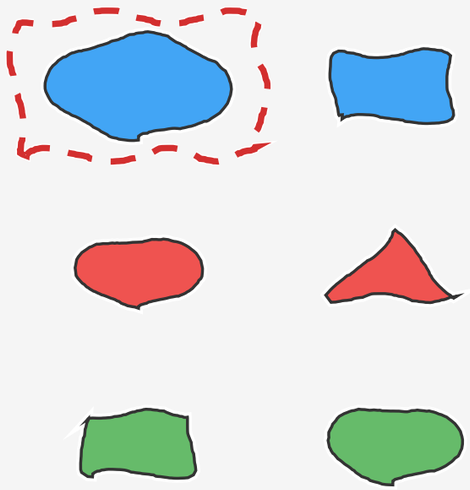
Visual Scene



"the big blue sticker"

target = dashed frame

RSA Model

Compositional Semantics

static: θ fixed from scene prior

recursive: θ updates with listener

word-by-word (incremental)

Literal Listener

L□(o | u)

Pragmatic Speaker

S□(u | o*)

full utterance (global)

a rationality · β LM-prior / b subj-bias

Exp 1: Slider Rating

"big blue" vs "blue big"

continuous preference (0–1)

ZOIB likelihood

0 — 1

Exp 2: Free Production

S · SC · CS · SCF · CSF · ...

15 utterance types

categorical likelihood

3 referential contexts × 2 size discriminability

2 × 2 Model Comparison (Bayesian PSIS-LOO)

	Static θ	Recursive θ
Global Speaker	S_G × θ_fix	S_G × θ_rec
Incremental Speaker	S_I × θ_fix better	**S_I × θ_rec** **best □**

Key Findings

Q2. Incremental > Global (both datasets)

Q3. Recursive semantics helps, but only when paired with incremental speaker

□ synergistic interaction

Q1. Composition alone yields baseline ordering effect (simulation)